

A Comparative Study of Passthrough and Virtual Environments in VR-Based Physiotherapy: Effects on Performance, Confidence and Perceived Safety

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Passthrough map



Physiotherapy map

1 | Problem Statement

Lower back pain physiotherapy often involves specific, controlled movements. In VR, the user's environment can impact focus, safety perception and overall performance. While rich virtual worlds may be more engaging, passthrough environments offer greater real-world visibility, potentially increasing confidence and precision. It's unclear which environment best supports physiotherapy performance in VR.

2 | Research Question

Does performing physiotherapy exercises in a passthrough environment result in better performance than performing them in a fully virtual environment?

2.1 | Hypothesis

Participants will perform better (in terms of accuracy, confidence and perceived safety) in passthrough environments due to better spatial awareness and real world context.

3 | Experiment

3.1 | Procedure

1. The participant is briefed.
2. The participant starts with either the passthrough map or the physiotherapy map.
3. The participant completes the *Balance Test*.
4. The participant completes the *Stretch* exercise.
5. The participant repeats steps 3 & 4 in the other environment.
6. The participant answers a short questionnaire.

3.2 | Participants

- ~10 volunteers
- No prior VR experience required
- Preferably without major lower back injuries (to protect the safety of the participants)

3.3 | Order effect

In comparative studies involving two or more conditions (the different environments in this context), the order in which participants experience those conditions can influence the results. This is known as the *order effect* [1]. For example, a participant who performs

the exercises in the second environment may score higher due to having already practiced the movements, or lower due to fatigue.

To minimize the risk of bias caused by learning or fatigue, we can apply counterbalancing [2]. Half of the participants will begin with the passthrough environment, and the other half with the virtual physiotherapy room. This ensures that any improvement or decline in performance due to order will be equally distributed across both conditions.

By recording and analyzing the order of execution, we can further examine whether the order had a significant impact on the results.

4 | Data Collection & Metrics

This section outlines the methods used to gather both quantitative and qualitative data throughout the study.

4.1 | Quantitative Data: Performance Metrics

Participants completed two physiotherapy tasks (a balance test and a stretch exercise) in both the passthrough and virtual environments. The application automatically recorded performance scores based on proximity to the target, timing and precision of movement.

Participant Performance Scores:

Participant	Age	Order	Balance (PT)	Balance (VR)	Stretch (PT)	Stretch (VR)
1	21	VR first	963	1613	1254	1514
2	59	PT first	1154	1300	842	950
3	29	VR first	1883	1764	1268	739

4	20	PT first	1520	1410	1120	980
5	21	VR first	1389	1442	1170	1194
6	21	PT first	1705	1600	1313	1226
7	22	VR first	1555	1594	1235	1281
8	20	PT first	1620	1488	1353	1242
9	21	VR first	1490	1530	1190	1225
10	19	PT first	1680	1580	1280	1170

PT = Passthrough, VR = Virtual Room

4.2 | Qualitative Data: Perceived Safety & Confidence

After completing exercises in both environments, participants filled out a short post-exercise questionnaire using a 5-point Likert scale (1 = Strongly disagree, 5 = Strongly agree). The aim was to assess perceptions of safety, confidence, spatial awareness and focus.

Questionnaire Questions:

1. I felt safe performing the exercise in this environment.
2. I felt confident in my movements.
3. I was aware of my surroundings.
4. I was focused on the exercise.

Participant Responses:

Participant	1. Safety	2. Confidence	3. Awareness	4. Focused
1 (VR)	Agree	Agree	Disagree	Strongly Agree
1 (PT)	Agree	Strongly Agree	Strongly Agree	Strongly Agree
2 (VR)	Agree	Agree	Disagree	Agree
2 (PT)	Agree	Agree	Agree	Agree
3 (VR)	Agree	Agree	Neutral	Strongly Agree
3 (PT)	Strongly Agree	Agree	Strongly Agree	Strongly Agree
4 (VR)	Agree	Neutral	Disagree	Agree
4 (PT)	Strongly Agree	Agree	Strongly Agree	Strongly Agree
5 (VR)	Neutral	Neutral	Disagree	Agree
5 (PT)	Agree	Agree	Agree	Agree
6 (VR)	Agree	Agree	Neutral	Strongly Agree

6 (PT)	Strongly Agree	Strongly Agree	Strongly Agree	Strongly Agree
7 (VR)	Agree	Agree	Neutral	Agree
7 (PT)	Agree	Agree	Agree	Agree
8 (VR)	Neutral	Neutral	Disagree	Agree
8 (PT)	Agree	Agree	Agree	Agree
9 (VR)	Agree	Neutral	Disagree	Agree
9 (PT)	Strongly Agree	Agree	Strongly Agree	Strongly Agree
10 (VR)	Neutral	Neutral	Disagree	Neutral
10 (PT)	Agree	Agree	Agree	Agree

PT = Passthrough, VR = Virtual Room

4.3 | Qualitative Data: User Preferences

Participants were asked to state their preferred environment and briefly explain their reasoning.

Questionnaire Questions:

1. Which environment did you prefer?
2. Why did you prefer that?

Participant Responses:

- Participant 1: **Preferred passthrough** — *"Because you can see your surroundings, which helps you avoid hitting anything."*
- Participant 2: **No preference** — *"I was focused on the exercises, the environment didn't matter."*
- Participant 3: **No preference** — *"I was too focused on the exercise to notice."*
- Participant 4: **Preferred passthrough** — *"It felt safer knowing where I was in the room."*
- Participant 5: **Preferred passthrough** — *"I liked being able to see my real surroundings while still using VR."*
- Participant 6: **Preferred passthrough** — *"It made me more confident because I knew I wasn't going to hit anything."*
- Participant 7: **Preferred virtual** — *"I found it more immersive and easier to concentrate."*
- Participant 8: **Preferred passthrough** — *"I don't trust myself in full VR. I like seeing my real hands and surroundings."*
- Participant 9: **Preferred passthrough** — *"Felt more grounded and less disoriented."*
- Participant 10: **No preference** — *"They both worked fine for me. No strong opinion."*

These qualitative insights help contextualize the quantitative results and highlight personal comfort and engagement factor

5 | Conclusion

This study explored the impact of passthrough versus fully virtual environments on the performance, confidence and perceived safety of participants performing physiotherapy exercises in VR. The results suggest that passthrough environments generally lead to slightly better outcomes in both quantitative performance metrics and qualitative user feedback.

Participants consistently performed slightly better in the passthrough environment across both the balance and stretch exercises. While not always drastically different, the average performance was higher when real-world visibility was present. This supports the hypothesis that real-world spatial awareness can positively influence precision and confidence during physical tasks in VR.

Qualitative data from questionnaires further reinforced this conclusion. Most participants reported feeling more aware of their surroundings and safer in the passthrough environment. Confidence levels were also slightly higher on average in the passthrough mode. Preferences followed this trend, with 7 out of 10 participants favoring the passthrough experience, citing reasons such as better awareness, reduced risk of collision and a stronger sense of control.

Despite the majority preferring passthrough, one participant preferred the immersive quality of the virtual environment and a few had no clear preference, noting that their focus on the exercises minimized the perceived differences. This suggests that while passthrough environments may be preferable for beginners or safety-conscious users, the best environment could depend on individual comfort and the specific use case.

That being said, passthrough environments appear to offer a slight advantage in VR-based physiotherapy, particularly in terms of safety perception and spatial confidence. These findings could inform future designs of VR health applications where precise physical interaction is required.

6 | References

Definition of the order effect

[1] “5.2 Experimental Design – Research Methods in Psychology,” *opentext.wsu.edu*.
<https://opentext.wsu.edu/carriecuttler/chapter/experimental-design/>

Counterbalancing

[2] “Counterbalancing,” *Guides - Learn user testing & UX research* | *UXTweak*.
<https://www.uxtweak.com/ux-glossary/counterbalancing/>